# Launch vehicle conceptual design report

TSTO-LOX-RP1-10t-LEO — 2 stages

| GLOM | PAYLOAD | ΔV REQUIRED | LENGTH |
|---|---|---|---|
| 410,003 kg | 10,000 kg | 9,408 m/s | 54.87 m |
| GROSS LIFT-OFF MASS | TO THE INJECTION ORBIT | INCL. LOSSES AND MARGIN | NOZZLE EXIT TO FAIRING TIP |

| MAX DIAMETER | LIFT-OFF T/W | PAYLOAD FRACTION |
|---|---|---|
| 4.00 m | 1.30 | 2.44 % |
| CORE / FAIRING | THRUST 5,227 KN | OF GLOM |

**Converged** in 23 iterations (relative residual 8.18e-10, tolerance 1e-09). Achieved Δv 9,408 m/s against a requirement of 9,408 m/s.

| Item | Value |
|---|---|
| Design | TSTO-LOX-RP1-10t-LEO |
| Configuration | 2 stages |
| Mission | LEO, payload 10,000 kg |
| Design file | `config/default_design.yaml` sha256:05b24045b70d37b9 |
| Assumptions file | `config/engineering_assumptions.yaml` sha256:7249f30091bdca40 |
| Tool version | launch_vehicle 0.4.0 |
| Generated | 2026-10-12 15:05 |
| Reference | Edberg & Costa, *Design of Rockets and Space Launch Vehicles*, 2nd ed., AIAA 2022 (with the published 1st-edition errata applied; see docs/ERRATA_COMPLIANCE.md) |

This report is a **conceptual sizing** result produced by mass-estimating relationships and closed-form geometry. It is not a design qualification and does not demonstrate that the vehicle can fly. In particular:

- **No trajectory simulation.** Ascent losses are either a published regression fit or user estimates; the achievability of the orbit is not demonstrated, and no steering or lofting profile exists.
- **No structural analysis.** Tank and structure masses come from statistical MERs (or thin-wall membrane theory when that option is selected). Buckling, axial and bending loads, slosh, acoustics and proof testing are not sized. Nothing here is a stress report.
- **No stability, control, thermal, GNC, separation or aerodynamic analysis.** Moments of inertia are reported, but no control authority or margin is assessed.
- **MER scatter.** Mass-estimating relationships are fits to historical vehicles; individual components can deviate by tens of per cent. Validation against twelve real vehicles gives a median GLOM error of about 17 % (see docs/VALIDATION_REPORT.md). Treat every mass here as having that kind of uncertainty.
- **Engines and motors are rubber.** Unless a real engine's mass and dimensions were entered, the engine is a scaled MER and its envelope is a layout estimate.

Every input used is listed in full, with its source and provenance, in the **Input register** section; every calculation is listed in the **Engineering method register** with the book page it comes from or an explicit mark that it is supplementary.

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## 1. Mission requirement and Δv budget

| Requirement | Value |
|---|---|
| Injection orbit | LEO: 400 km circular, i = 28.50 deg |
| Launch site | latitude 28.50 deg, altitude 0 m |
| Launch azimuth | computed from the inclination |
| Earth rotation credited | yes |
| Loss model | book_penalty — Eqs. 3:35-3:38 regression (lumped) |
| Design margin | 0.0 % |
| Air launch | no (ground launch) |
| Δv override | none |
| Δv allocation | fractions → stage 1: 4,139.4 m/s, stage 2: 5,268.3 m/s |

### Δv budget, line by line

| Budget line | m/s | Category | In total | Source |
|---|---|---|---|---|
| Ideal injection speed (inertial, at perigee) | 7,671.7 | ideal | yes | Eq. 3:1 / 3:4 (book pp.156-157) |
| Earth-rotation contribution (vector, launch site) | -408.7 | rotation | yes | Book Example 3.9 (p.185) method; negative = benefit |
| Ascent losses: gravity + drag + steering + propulsion (lumped, Book Eq. 3:37-3:38) | 2,144.6 | loss | yes | Book regression pp.181-182: K3=1266.5, K4=1.9405, T_mix=452.5 s (T_ascent=548.4 s, T_3s=387.2 s) |
| indicative gravity share of lumped losses (Table 3.2 proportions) *(informational)* | 1,784.6 | info | no | Informational split only - NOT added to the total (already inside the lumped penalty) |
| indicative drag share of lumped losses (Table 3.2 proportions) *(informational)* | 135.6 | info | no | Informational split only - NOT added to the total (already inside the lumped penalty) |
| indicative steering share of lumped losses (Table 3.2 proportions) *(informational)* | 134.4 | info | no | Informational split only - NOT added to the total (already inside the lumped penalty) |
| indicative propulsion share of lumped losses (Table 3.2 proportions) *(informational)* | 90.0 | info | no | Informational split only - NOT added to the total (already inside the lumped penalty) |
| Other allowances (reserves, extra) | 0.0 | allowance | yes | User-defined |
| Design margin (0.0% of subtotal) | 0.0 | margin | yes | User-defined fraction applied once to the subtotal |
| **Δv required** | **9,407.6** | **total** | | |

## 2. Vehicle configuration

| Stage | Name | Type | Propellant | Propulsion | D [m] | L [m] | Isp_eff [s] | Δv [m/s] | Propellant [kg] | Dry [kg] | s = ms/(ms+mp) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Stage 1 | liquid | LOX/RP-1, O/F 2.50 | 6 x Generic LOX/RP-1 booster engine | 4.00 | 33.88 | 290.0 | 4,139 | 322,638 | 14,156 | 0.0656 |
| 2 | Stage 2 | liquid | LOX/RP-1, O/F 2.50 | 1 x Generic LOX/RP-1 vacuum engine | 3.00 | 12.92 | 340.0 | 5,268 | 59,037 | 2,661 | 0.0666 |

| Upper assembly | Value |
|---|---|
| Payload | 10,000 kg in a 2.70 m x 3.00 m envelope |
| Payload adapter (PAF) | 805.0 kg (method: mer) |
| Fairing | 705.3 kg, 3.00 m diameter, metal, jettisoned at the start of stage 2 |

## 3. Drawings

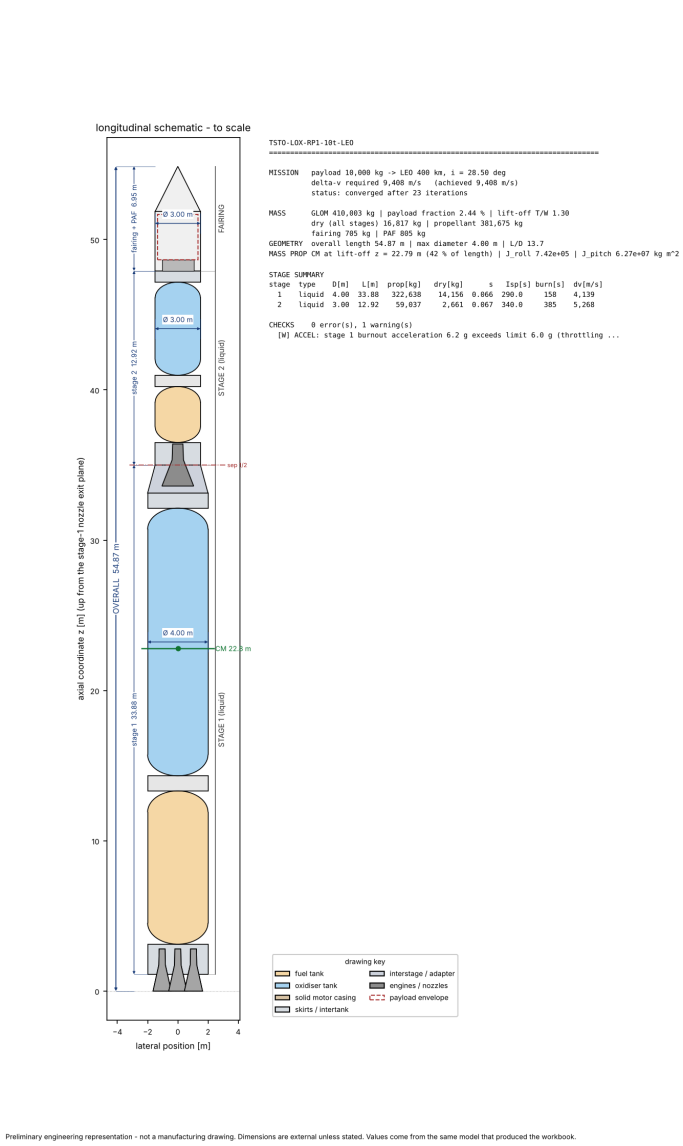



**Figure 1.** Vehicle general arrangement, to scale. Dimensions are the sized values; the right-hand panel lists the numbers for each stage. Interior tank outlines are the sized tank or motor-case geometry, not a structural layout.

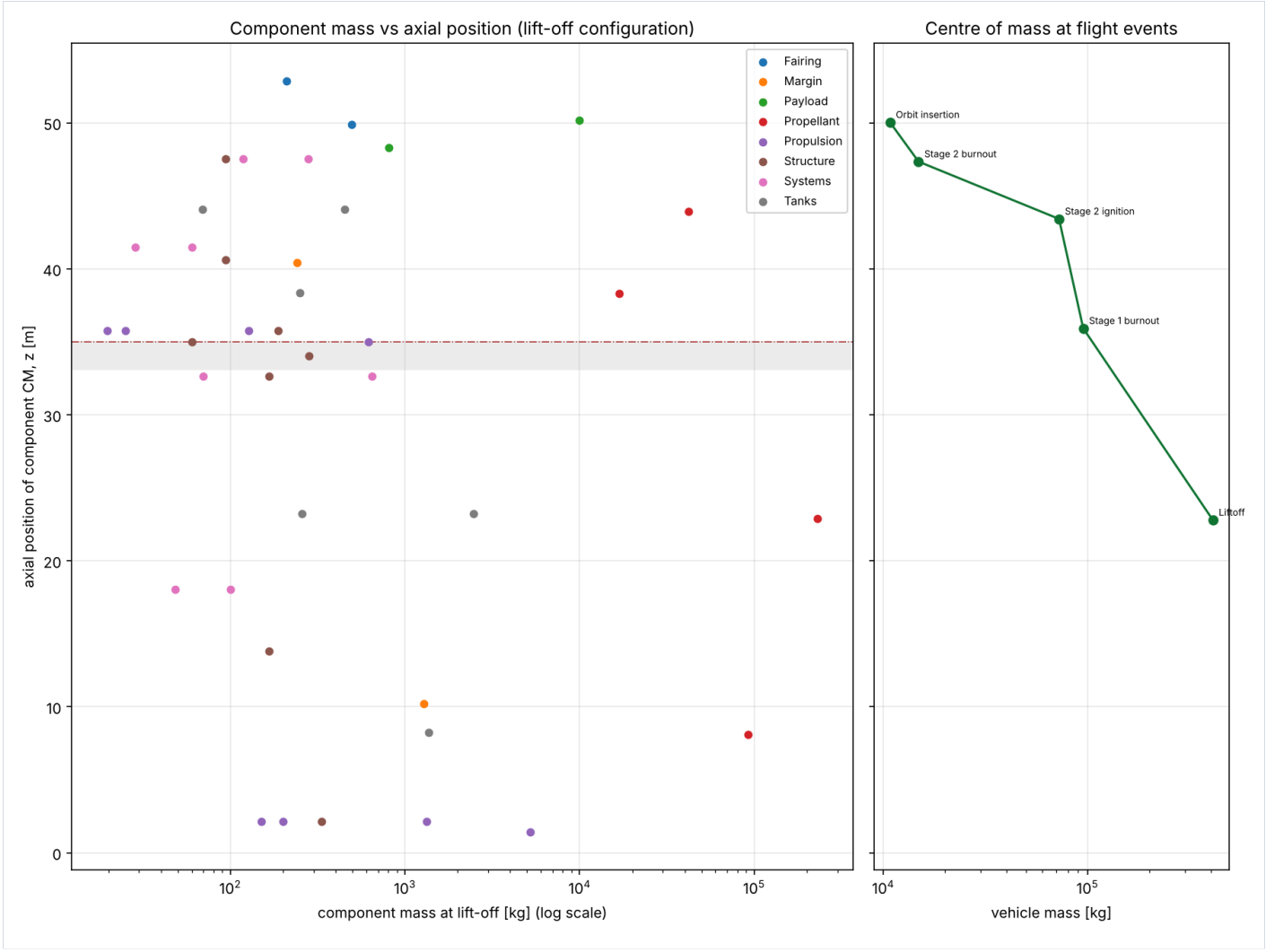



**Figure 2.** Mass distribution along the vehicle axis and the centre-of-mass travel through the flight events, from the mass ledger in section 6.

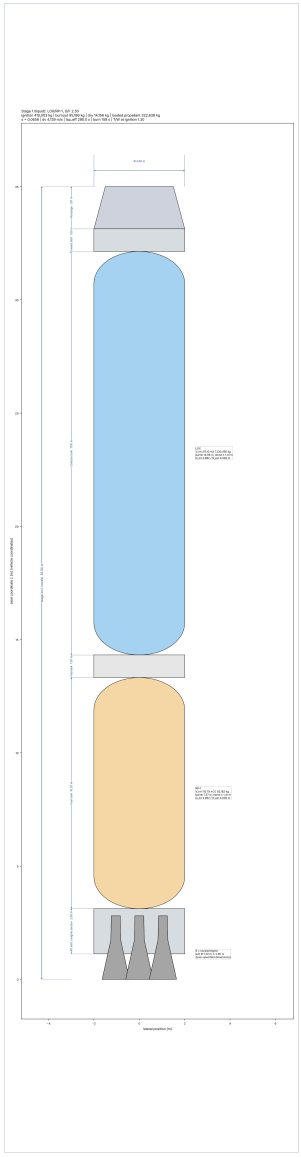



**Figure 3.** Stage 1 — tankage and engines, to scale.

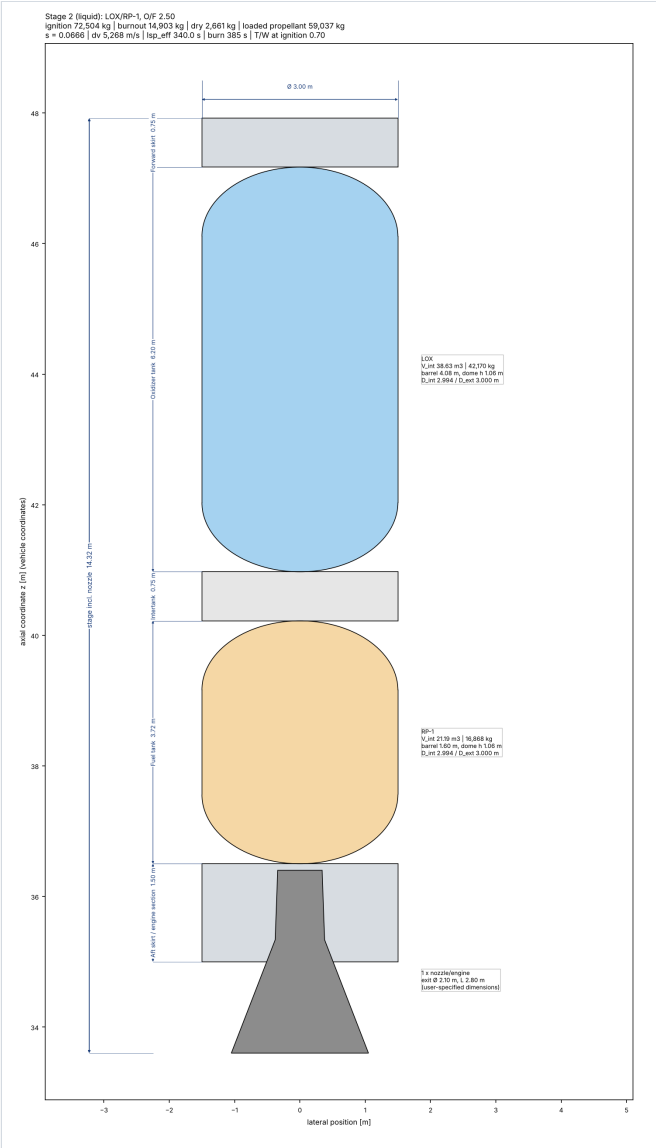



**Figure 4.** Stage 2 — tankage and engines, to scale.

## 4. Propulsion and performance

| Stage | T_vac [kN] | T_SL [kN] | Reference condition | mdot [kg/s] | t_burn [s] | T/W ignition | T/W burnout | a_max [g] | mass ratio μ | Δv recomputed [m/s] |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 5,807.8 | 5,227.0 | sea_level | 1,974.1 | 158.5 | 1.30 | 6.22 | 6.22 | 4.2867 | 4,139.4 |
| 2 | 497.7 | - | vacuum | 149.3 | 384.9 | 0.70 | 3.41 | 3.41 | 4.8552 | 5,268.3 |

The last column recomputes Δv from the sized burn-start and burnout masses with the rocket equation; it must match the allocated Δv in section 2 to solver tolerance.

## 5. Mass statement

| Stage | Carried above [kg] | Mission prop. [kg] | Start-up [kg] | Residual [kg] | Loading margin [kg] | Loaded prop. [kg] | Dry [kg] | Inert at burnout [kg] | Ignition [kg] | Burnout [kg] |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 73,209 | 312,843 | 1,974.1 | 6,256.9 | 1,564.2 | 322,638 | 14,156 | 21,977 | 410,003 | 95,186 |
| 2 | 10,805 | 57,452 | 149.3 | 1,149.0 | 287.3 | 59,037 | 2,661 | 4,098 | 72,504 | 14,903 |
| **Stack** | | **370,295** | **2,123.4** | **7,405.9** | **1,851.5** | **381,675** | **16,817** | | **410,003** | |

"Carried above" is everything above the stage at its ignition — payload, adapter, the fairing while it is still attached, and every upper stage fully loaded. "Inert at burnout" is dry + residual + loading margin, the mass the structural factor s uses.

### GLOM build-up

| Item | Mass [kg] |
|---|---|
| Payload | 10,000 |
| Payload adapter | 805.0 |
| Fairing | 705.3 |
| Stage 1 dry | 14,156 |
| Stage 1 propellant loaded | 322,638 |

| Item | Mass [kg] |
|---|---|
| Stage 2 dry | 2,661 |
| Stage 2 propellant loaded | 59,037 |
| **GLOM** | **410,003** |
| Closure against the sized GLOM | +0.000 kg |

### Dry-mass breakdown by component

#### Stage 1

| Component | Mass [kg] | % of dry |
|---|---|---|
| Engines | 5,224.0 | 36.9 |
| Tank structure (oxidizer) | 2,465.9 | 17.4 |
| Tank structure (fuel) | 1,364.3 | 9.6 |
| Thrust structure | 1,332.9 | 9.4 |
| Mass growth margin | 1,286.9 | 9.1 |
| Pressurant system | 645.3 | 4.6 |
| Aft skirt | 334.3 | 2.4 |
| Interstage | 282.5 | 2.0 |
| Tank insulation (oxidizer) | 257.3 | 1.8 |
| Engine gimbals | 199.8 | 1.4 |
| Intertank | 167.1 | 1.2 |
| Forward skirt | 167.1 | 1.2 |
| Feed system | 150.0 | 1.1 |
| Other equipment | 100.0 | 0.7 |
| Avionics | 70.0 | 0.5 |
| Separation hardware | 60.0 | 0.4 |
| Wiring | 48.5 | 0.3 |
| **Stage dry mass** | **14,155.8** | **100.0** |

#### Stage 2

| Component | Mass [kg] | % of dry |
|---|---|---|
| Engines | 615.4 | 23.1 |
| Tank structure (oxidizer) | 451.2 | 17.0 |
| Avionics | 280.0 | 10.5 |
| Tank structure (fuel) | 249.6 | 9.4 |
| Mass growth margin | 241.9 | 9.1 |
| Aft skirt | 188.0 | 7.1 |
| Thrust structure | 126.9 | 4.8 |
| Pressurant system | 118.1 | 4.4 |
| Intertank | 94.0 | 3.5 |
| Forward skirt | 94.0 | 3.5 |
| Tank insulation (oxidizer) | 68.9 | 2.6 |
| Other equipment | 60.0 | 2.3 |
| Wiring | 28.4 | 1.1 |
| Feed system | 25.0 | 0.9 |
| Engine gimbals | 19.7 | 0.7 |
| **Stage dry mass** | **2,661.3** | **100.0** |

## 6. Mass ledger and mass properties

Every mass in the vehicle, where it sits on the axis, which shape it was given for inertia, and the rule it came from. This table is the input to the centre-of-mass and inertia calculation and to Figure 2.

▼ **Full mass ledger (44 items)**

| Stage | Group | Component | Mass [kg] | z_cm [m] | Kind | Inertia shape | Source / assumption |
|---|---|---|---|---|---|---|---|
| 1 | Propulsion | Engines | 5,224.0 | 1.400 | dry | point masses on engine ring (Table 8.15: point mass) | Table 8.9 MER; mid-length of engine envelope |
| 1 | Propulsion | Engine gimbals | 199.8 | 2.120 | dry | point masses on engine ring | MER; aft skirt mid |
| 1 | Propulsion | Thrust structure | 1,332.9 | 2.120 | dry | thin cylinder r=R/2, h=D/2 (assumed) | MER; aft skirt mid |

| Stage | Group | Component | Mass [kg] | z_cm [m] | Kind | Inertia shape | Source / assumption |
|---|---|---|---|---|---|---|---|
| 1 | Tanks | Tank structure (fuel) | 1,364.3 | 8.220 | dry | thin shell of revolution (numerical) | tank method per tank spec; shell CM from outline integration |
| 1 | Tanks | Tank structure (oxidizer) | 2,465.9 | 23.228 | dry | thin shell of revolution (numerical) | tank method per tank spec; shell CM from outline integration |
| 1 | Tanks | Tank insulation (oxidizer) | 257.3 | 23.228 | dry | thin shell of revolution (numerical) | tank method per tank spec; shell CM from outline integration |
| 1 | Structure | Aft skirt | 334.3 | 2.120 | dry | thin cylinder (Table 8.10) | 13.3 kg/m^2 x area (Table 8.8); thin cylinder (Table 8.10) |
| 1 | Structure | Intertank | 167.1 | 13.820 | dry | thin cylinder (Table 8.10) | 13.3 kg/m^2 x area (Table 8.8); thin cylinder (Table 8.10) |
| 1 | Structure | Forward skirt | 167.1 | 32.635 | dry | thin cylinder (Table 8.10) | 13.3 kg/m^2 x area (Table 8.8); thin cylinder (Table 8.10) |
| 1 | Structure | Interstage | 282.5 | 34.024 | dry | thin frustum (Table 8.10) | 13.3 kg/m^2 x frustum area (Table 8.8); CM per Table 8.10 |
| 1 | Systems | Avionics | 70.0 | 32.635 | dry | thin ring (Table 8.15/8.16) | MER/assumption; forward skirt mid |
| 1 | Systems | Wiring | 48.5 | 18.061 | dry | thin bar along stage (Table 8.17) | 1.43 kg/m; mid-stage (Table 8.15: 1/2 l) |
| 1 | Propulsion | Feed system | 150.0 | 2.120 | dry | point (on axis) | assumption; aft skirt mid |
| 1 | Systems | Pressurant system | 645.3 | 32.635 | dry | point (on axis) | assumption; forward skirt mid |
| 1 | Structure | Separation hardware | 60.0 | 35.001 | dry | thin ring | assumption; top of stage |
| 1 | Systems | Other equipment | 100.0 | 18.061 | dry | point (on axis) | assumption; mid-stage |
| 1 | Margin | Mass growth margin | 1,286.9 | 10.223 | dry | scaled copy of stage components | single margin; distributed like the stage's other dry components |
| 1 | Propellant | Propellant loaded (fuel, RP-1) | 92,182.3 | 8.080 | prop_loaded | liquid solid of revolution (numerical) | liquid column centroid, full load |
| 1 | Propellant | Residual + margin propellant (fuel) | 2,234.6 | 3.515 | prop_resid | liquid solid of revolution (numerical) | liquid at tank bottom at burnout |
| 1 | Propellant | Propellant loaded (oxidizer, LOX) | 230,455.7 | 22.867 | prop_loaded | liquid solid of revolution (numerical) | liquid column centroid, full load |
| 1 | Propellant | Residual + margin propellant (oxidizer) | 5,586.5 | 14.860 | prop_resid | liquid solid of revolution (numerical) | liquid at tank bottom at burnout |
| 2 | Propulsion | Engines | 615.4 | 35.001 | dry | point masses on engine ring (Table 8.15: point mass) | Table 8.9 MER; mid-length of engine envelope |
| 2 | Propulsion | Engine gimbals | 19.7 | 35.751 | dry | point masses on engine ring | MER; aft skirt mid |
| 2 | Propulsion | Thrust structure | 126.9 | 35.751 | dry | thin cylinder r=R/2, h=D/2 (assumed) | MER; aft skirt mid |
| 2 | Tanks | Tank structure (fuel) | 249.6 | 38.362 | dry | thin shell of revolution (numerical) | tank method per tank spec; shell CM from outline integration |
| 2 | Tanks | Tank structure (oxidizer) | 451.2 | 44.071 | dry | thin shell of revolution (numerical) | tank method per tank spec; shell CM from outline integration |
| 2 | Tanks | Tank insulation (oxidizer) | 68.9 | 44.071 | dry | thin shell of revolution (numerical) | tank method per tank spec; shell CM from outline integration |
| 2 | Structure | Aft skirt | 188.0 | 35.751 | dry | thin cylinder (Table 8.10) | 13.3 kg/m^2 x area (Table 8.8); thin cylinder (Table 8.10) |
| 2 | Structure | Intertank | 94.0 | 40.597 | dry | thin cylinder (Table 8.10) | 13.3 kg/m^2 x area (Table 8.8); thin cylinder (Table 8.10) |
| 2 | Structure | Forward skirt | 94.0 | 47.546 | dry | thin cylinder (Table 8.10) | 13.3 kg/m^2 x area (Table 8.8); thin cylinder (Table 8.10) |
| 2 | Systems | Avionics | 280.0 | 47.546 | dry | thin ring (Table 8.15/8.16) | MER/assumption; forward skirt mid |
| 2 | Systems | Wiring | 28.4 | 41.461 | dry | thin bar along stage (Table 8.17) | 1.43 kg/m; mid-stage (Table 8.15: 1/2 l) |
| 2 | Propulsion | Feed system | 25.0 | 35.751 | dry | point (on axis) | assumption; aft skirt mid |
| 2 | Systems | Pressurant system | 118.1 | 47.546 | dry | point (on axis) | assumption; forward skirt mid |
| 2 | Systems | Other equipment | 60.0 | 41.461 | dry | point (on axis) | assumption; mid-stage |
| 2 | Margin | Mass growth margin | 241.9 | 40.414 | dry | scaled copy of stage components | single margin; distributed like the stage's other dry components |
| 2 | Propellant | Propellant loaded (fuel, RP-1) | 16,867.8 | 38.312 | prop_loaded | liquid solid of revolution (numerical) | liquid column centroid, full load |
| 2 | Propellant | Residual + margin propellant (fuel) | 410.4 | 36.694 | prop_resid | liquid solid of revolution (numerical) | liquid at tank bottom at burnout |
| 2 | Propellant | Propellant loaded (oxidizer, LOX) | 42,169.5 | 43.950 | prop_loaded | liquid solid of revolution (numerical) | liquid column centroid, full load |
| 2 | Propellant | Residual + margin propellant (oxidizer) | 1,025.9 | 41.233 | prop_resid | liquid solid of revolution (numerical) | liquid at tank bottom at burnout |
| 0 | Payload | Payload | 10,000.0 | 50.171 | payload | solid cylinder (Table 8.17), uniform density assumed | envelope mid-length; uniform solid cylinder assumed |
| 0 | Payload | Payload adapter (PAF) | 805.0 | 48.296 | paf | thin cylinder (assumed) | PAF mid-length |
| upper | Fairing | Payload fairing (cylinder) | 495.1 | 49.896 | fairing | thin cylinder (Table 8.10) | area-weighted share; thin cylinder (Table 8.10) |

| Stage | Group | Component | Mass [kg] | z_cm [m] | Kind | Inertia shape | Source / assumption |
|---|---|---|---|---|---|---|---|
| upper | Fairing | Payload fairing (nose cone) | 210.2 | 52.871 | fairing | thin cone (Table 8.10) | area-weighted share; thin cone, CM at h/3 (Table 8.10) |
| | | **Total** | **419,260** | | | | |

### Centre of mass and inertia through the flight events

| Event | Mass [kg] | z_cm [m] | z_cm/L | J_roll [kg m2] | J_pitch [kg m2] | J_roll/J_pitch | Note |
|---|---|---|---|---|---|---|---|
| Liftoff (stage 1 ignition, full load) | 410,003 | 22.788 | 0.415 | 742,046 | 62,654,842 | 0.0118 | |
| Stage 1 burnout (before separation) | 95,186 | 35.921 | 0.655 | 118,455 | 21,517,929 | 0.0055 | |
| Stage 2 ignition (after stage 1 separation, fairing jettisoned) | 72,504 | 43.415 | 0.791 | 77,330 | 1,175,855 | 0.0658 | |
| Stage 2 burnout (before separation) | 14,903 | 47.341 | 0.863 | 14,848 | 375,908 | 0.0395 | |
| Orbit insertion (payload + adapter) | 10,805 | 50.031 | 0.912 | 10,924 | 15,619 | 0.6994 | |

Component locations/shapes are preliminary lumped assumptions (see ledger 'shape' column). Inertias use the book's parallel-axis method; thrust structure, feed system, payload and PAF shapes are assumptions; products of inertia are zero by the axisymmetry assumption.

## 7. Geometry

▼ Segment table (13 segments, bottom to top)

| Stage | Kind | Component | z bottom [m] | z top [m] | Length [m] | D bottom [m] | D top [m] |
|---|---|---|---|---|---|---|---|
| 1 | aft_skirt | Aft skirt / engine section 1 | 1.120 | 3.120 | 2.000 | 4.000 | 4.000 |
| 1 | tank | Fuel tank 1 | 3.120 | 13.320 | 10.200 | 4.000 | 4.000 |
| 1 | intertank | Intertank 1 | 13.320 | 14.320 | 1.000 | 4.000 | 4.000 |
| 1 | tank | Oxidizer tank 1 | 14.320 | 32.135 | 17.815 | 4.000 | 4.000 |
| 1 | fwd_skirt | Forward skirt 1 | 32.135 | 33.135 | 1.000 | 4.000 | 4.000 |
| 1 | interstage | Interstage 1-2 | 33.135 | 35.001 | 1.866 | 4.000 | 3.000 |
| 2 | aft_skirt | Aft skirt / engine section 2 | 35.001 | 36.501 | 1.500 | 3.000 | 3.000 |
| 2 | tank | Fuel tank 2 | 36.501 | 40.222 | 3.721 | 3.000 | 3.000 |
| 2 | intertank | Intertank 2 | 40.222 | 40.972 | 0.750 | 3.000 | 3.000 |
| 2 | tank | Oxidizer tank 2 | 40.972 | 47.171 | 6.198 | 3.000 | 3.000 |
| 2 | fwd_skirt | Forward skirt 2 | 47.171 | 47.921 | 0.750 | 3.000 | 3.000 |
| - | fairing_cyl | Fairing cylinder | 47.921 | 51.871 | 3.950 | 3.000 | 3.000 |
| - | fairing_nose | Fairing nose cone | 51.871 | 54.871 | 3.000 | 3.000 | 0.000 |

### Tank / motor-case geometry (book Eqs. 8:13-8:16, Table 8.2; solids Eqs. 8:23-8:24)

| Stage | Role | Propellant | V internal [m3] | V liquid [m3] | D internal [m] | Dome AR | Barrel L [m] | Dome h [m] | External L [m] | External area [m2] | Tank mass [kg] | Insulation |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | fuel | RP-1 | 115.790 | 112.417 | 3.992 | 1.414 | 7.369 | 1.411 | 10.200 | 133.40 | 1,364.3 | |
| 1 | oxidizer | LOX | 211.098 | 202.154 | 3.992 | 1.414 | 14.984 | 1.411 | 17.815 | 229.09 | 2,465.9 | 2 |
| 2 | fuel | RP-1 | 21.188 | 20.571 | 2.994 | 1.414 | 1.598 | 1.059 | 3.721 | 38.01 | 249.6 | |
| 2 | oxidizer | LOX | 38.627 | 36.991 | 2.994 | 1.414 | 4.075 | 1.059 | 6.198 | 61.36 | 451.2 | |

| Overall | Value |
|---|---|
| Overall length | 54.871 m |
| Maximum diameter | 4.000 m |
| Maximum span (incl. boosters) | 4.000 m |
| Length / diameter | 13.72 |
| Payload envelope | z = 48.671 to 51.671 m |

## 8. Input register

Every input the model read, with its units, what it means, where the value came from, and how it is classified. Nothing else was used; nothing here was tuned to make a result match a real vehicle.

user — a requirement or a choice the designer made  reference — taken from the book or a cited source  assumed — a placeholder that should be replaced by real data

86 assumed, 105 reference, 38 user, 229 inputs in total.

## 9. Checks and warnings

| Level | Code | Message |
|---|---|---|
| WARNING | ACCEL | stage 1 burnout acceleration 6.2 g exceeds limit 6.0 g (throttling not modelled) |

| Level | Code | Message |
|---|---|---|
| INFO | FIDELITY | Preliminary model: losses are user-estimated; no trajectory simulation, loads, stability or thermal analysis. Achievability of the orbit is NOT demonstrated by this tool. |

0 errors, 1 warnings. Warnings are not failures: they mark places where the model is being used near or outside the range its methods were fitted for.

## 10. Engineering method register

Every calculation in the tool, its formulation, the verified book page it comes from (printed page numbers; "SUPPLEMENTARY" means it is **not** claimed to be in the book), its assumptions and limits, where it is implemented, and how it is verified.